COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	02/09/26	**DATE SUBMITTED:**	02/09/26
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ACTIVITY NO. 7.2 — **Algorithm Analysis: Best-Case and Worst-Case Time Complexity**

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Finding an Element

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Best: O(1)
Worst: O(n)
Why: Found in first or last

2. Printing All Elements

```
for (int i = 0; i < n; i++)
{
    cout << arr[i] << endl;
}
```

Best: O(n)
Worst: O(n)
Why: n loop

3. Nested Loops

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

Best: O(n2)
Worst: O(n2)
Why: 2 loops and both at n times

4. Conditional Processing

```
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
else
{
    cout << "Small input";
}
```

Best: O(1)
Worst: O(n)
Why: If n is small = 1, but if n = big, it will run n times

5. Fixed Inner Loop

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
```

Best: O(n)
Worst: O(n)
Why: First loop is n times and 2nd loop runs 10 times

6. Two Consecutive Loops

```
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}

for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

Best: O(n)
Worst: O(n)
Why: Both is n times

7. Dividing by Two

```
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

Best: O(log n)
Worst: O(log n)
Why: Dividing 2 repeatedly

8. Nested Loop With Different Limits

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
```

Best: O(n2)
Worst: O(n2)
Why: The loop gets longer as it increase and it becomes n2

9. Early Termination

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
```

Best:O(1)
Worst: O(n)
Why: Either at first or at last

10. Two-Level Search

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
    }
}
```

Best: O(n)
Worst: O(n2)
Why: Because the inner loop still stops at the moment and the other one still runs at n times. And worst if both runs at n times

11. Linear Loop With Logarithmic Inner Work

```
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best: O(n log n)
Worst: O(n log n)
Why: Outer runs n time and inner logs it

12. Conditional Nested Loop

```
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

Best: O(1)
Worst: O(n2)
Why: Might be correct immediately or both might run n times

13. Finding the Maximum

```
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Best: O(n)
Worst: O(n)
Why: Checks everyhing

14. Three Nested Loops

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Best: O(n3)
Worst: O(n3)
Why: 3 loops running at n times

15. Comprehensive Analysis

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }

    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best: O(1)
Worst: O(n log n)
Why: Might stop immediately or it will run and take log n times

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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QUEZON CITY

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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